1. Multiplying by 5

To multiply any number by 5, divide the number by 2 and then multiply by 10 (i.e., add a zero at the end).

- **Trick**: $N \times 5 = (N/2) \times 10$
- **Example**: 128×5
  - Divide 128 by 2: $128/2 = 64$
  - Multiply by 10: $64 \times 10 = 640$
  - So, $128 \times 5 = 640$
- **Example**: 375×5
  - Divide 375 by 2: $375/2 = 187.5$
  - Multiply by 10: $187.5 \times 10 = 1875$
  - So, $375 \times 5 = 1875$

2. Multiplying by 15, 25, 35, etc. (Numbers ending in 5)

To multiply a number by $N5$(where $N$is the digit(s) before 5), you can use a variation of the trick for multiplying by 5.

- **Trick**: $X \times N5 = X \times (N \times 10 + 5) = (X \times N \times 10) + (X \times 5)$
  Alternatively, multiply by $N$and add half of the number to it, then append 25. If the number is odd, add 0.5 before appending 25 (effectively adding 12.5, which is $X/2 * 10$+ 5 with 25).
  A simpler rule for $X \times N5$: $(X \times N) \times 10 + X \times 5$
- **Example (Multiplying by 15)**: 124×15
  - $124 \times 15 = 124 \times (10 + 5) = (124 \times 10) + (124 \times 5)$
  - $124 \times 10 = 1240$
  - $124 \times 5 = 620$(using the previous trick)
  - $1240 + 620 = 1860$
  - So, $124 \times 15 = 1860$
- **Example (Multiplying by 25)**: 76×25
  - $76 \times 25 = 76 \times (20 + 5) = (76 \times 20) + (76 \times 5)$
  - $76 \times 20 = 1520$
  - $76 \times 5 = 380$

   - $1520 + 380 = 1900$
   - So, $76 \times 25 = 1900$
   - **Shortcut for 25**: $X \times 25 = (X/4) \times 100$. If $X$is not divisible by 4, perform division with remainder. $76/4 = 19$. $19 \times 100 = 1900$.

3. Multiplying by Numbers Consisting of Nines (9, 99, 999, ...)

To multiply a number by a sequence of 9s, subtract 1 from the number and then subtract the result from the original number of 9s.

- **Trick**: $N \times 99...9 = (N-1)$ followed by $(100...0 - N)$
  The number of zeros in $100...0$should match the number of 9s.
- **Example**: 456×99
   - Subtract 1 from 456: $456 - 1 = 455$
   - Subtract 456 from 1000 (since 99 has two 9s, use 10^2): $1000 - 456 = 544$
   - Combine: $455$and 544give 45544
   - So, $456 \times 99 = 45544$
- **Example**: 987×999
   - Subtract 1 from 987: $987 - 1 = 986$
   - Subtract 987 from 1000: $1000 - 987 = 13$
   - Combine (ensure the second part has same number of digits as 9s): $986$and 013give 986013
   - So, $987 \times 999 = 986013$

4. Multiplying by 11

This trick involves a simple addition process.

- **Trick**: For a 2-digit number AB, the product is $A(A+B)B$. If $A+B$is a 2-digit number, carry over the tens digit.
- **Example**: 34×11
   - Write down the last digit: 4
   - Add the digits: $3 + 4 = 7$
   - Write down the first digit: 3
   - Result: 374

- **Example**: 78×11
  - Write down the last digit: 8
  - Add the digits: $7 + 8 = 15$. Write down 5and carry over 1.
  - Write down the first digit and add the carry-over: $7 + 1 = 8$
  - Result: 858
- **For 3-digit numbers** $ABC: A(A + B)(B + C)C$. Carry over as needed.
  - **Example**: 123×11
    - $1(1 + 2)(2 + 3)3ightarrow1353$
  - **Example**: 456×11
    - $4(4 + 5)(5 + 6)6ightarrow49(11)6$. Carry over the 1 from 11 to 9: $4(9 + 1)16ightarrow5016$
    - Result: 5016

5. Multiplying Numbers Close to a Power of 10 (e.g., 98 x 97)

This method is efficient when both numbers are slightly less than a power of 10 (like 100).

- **Trick**: $(100 - a) \times (100 - b) = 100(100 - (a + b)) + ab$
  The result is in two parts: the left part is $100 - (a + b)$and the right part is $a \times b$. Ensure the right part has the same number of digits as the power of 10 (e.g., 2 digits for 100).
- **Example**: 98×97
  - Both numbers are less than 100.
  - 98is $100 - 2$(so $a = 2$)
  - 97is $100 - 3$(so $b = 3$)
  - Left part: $100 - (a + b) = 100 - (2 + 3) = 100 - 5 = 95$
  - Right part: $a \times b = 2 \times 3 = 6$. Pad with a zero to make it 2 digits: 06
  - Combine: 9506
  - So, $98 \times 97 = 9506$
- **Example**: 91×89
  - 91is $100 - 9(a = 9)$

- 89 is $100 - 11 (b = 11)$
- Left part: $100 - (9 + 11) = 100 - 20 = 80$
- Right part: $9 \times 11 = 99$
- Combine: 8099
- So, $91 \times 89 = 8099$

- **Trick for numbers above 100**: $(100 + a) \times (100 + b) = 100(100 + (a + b)) + ab$
    - **Example**: 102×105
        - $102 = 100 + 2 (a = 2)$
        - $105 = 100 + 5 (b = 5)$
        - Left part: $100 + (2 + 5) = 107$
        - Right part: $2 \times 5 = 10$
        - Combine: 10710
        - So, $102 \times 105 = 10710$

6. Multiplying Numbers with Same Tens Digit and Units Digits Summing to 10 (e.g., 23 x 27)

This is a very popular and quick trick.

- **Trick**: For numbers like $AB \times AC$ where $B + C = 10$, the product is $(A \times (A + 1))$ followed by $(B \times C)$.
- **Example**: 23×27
    - Tens digit is 2 for both. Units digits are 3 and 7, and $3 + 7 = 10$.
    - Left part: Multiply the tens digit by one more than itself: $2 \times (2 + 1) = 2 \times 3 = 6$
    - Right part: Multiply the units digits: $3 \times 7 = 21$
    - Combine: 6 followed by 21 gives 621
    - So, $23 \times 27 = 621$
- **Example**: 84×86
    - Tens digit is 8. Units digits are 4 and 6, and $4 + 6 = 10$.
    - Left part: $8 \times (8 + 1) = 8 \times 9 = 72$

  - Right part: $4 \times 6 = 24$
  - Combine: 7224
  - So, $84 \times 86 = 7224$

7. General Multiplication (Vertical and Crosswise - Nikhilam Sutra)

This is a versatile method for multiplying any two numbers, irrespective of their size.

- **Trick**: For $AB \times CD$(where $A, C$are tens digits and $B, D$are units digits).
  The method has three steps:
  1. Rightmost digit: Multiply the units digits ($B \times D$). Write down the last digit and carry over the rest.
  2. Middle part: Cross-multiply and add ($A \times D + C \times B$) and add any carry-over from step 1. Write down the last digit and carry over the rest.
  3. Leftmost part: Multiply the tens digits ($A \times C$) and add any carry-over from step 2. Write down the result.
- **Example**: 23×45
  - Units digits: $3 \times 5 = 15$. Write down 5, carry over 1.
  - Cross-multiplication: $(2 \times 5) + (4 \times 3) = 10 + 12 = 22$. Add carry-over 1: $22 + 1 = 23$. Write down 3, carry over 2.
  - Tens digits: $2 \times 4 = 8$. Add carry-over 2: $8 + 2 = 10$. Write down 10.
  - Result: 1035
  - So, $23 \times 45 = 1035$
- **Example**: 123×456(for 3-digit numbers)
  The process is similar but involves more steps:
  1. Units: $3 \times 6 = 18$. Write 8, carry 1.
  2. Tens: $(2 \times 6) + (3 \times 5) + 1(extcarry) = 12 + 15 + 1 = 28$. Write 8, carry 2.
  3. Hundreds: $(1 \times 6) + (3 \times 4) + (2 \times 5) + 2(extcarry) = 6 + 12 + 10 + 2 = 30$. Write 0, carry 3.
  4. Thousands: $(1 \times 5) + (2 \times 4) + 3(extcarry) = 5 + 8 + 3 = 16$. Write 6, carry 1.
  5. Ten Thousands: $(1 \times 4) + 1(extcarry) = 4 + 1 = 5$. Write 5.
  - Result: 56088

- So, $123 \times 456 = 56088$

8. Multiplying by 111, 1111, etc.

Similar to multiplying by 11, but involves summing adjacent digits in groups.

- **Trick**: For $N \times 111$:
    - Sum of single digits.
    - Sum of pairs of adjacent digits.
    - Sum of triplets of adjacent digits.
    - Then work backwards, carrying over.
- **Example**: 234×111
    - Rightmost digit: 4
    - Next: $3 + 4 = 7$
    - Next: $2 + 3 + 4 = 9$
    - Next: $2 + 3 = 5$
    - Leftmost digit: 2
    - Result: 25974
- **Example**: 567×111
    - Rightmost: 7
    - $6 + 7 = 13$. Write 3, carry 1.
    - $5 + 6 + 7 = 18$. Add carry 1: $18 + 1 = 19$. Write 9, carry 1.
    - $5 + 6 = 11$. Add carry 1: $11 + 1 = 12$. Write 2, carry 1.
    - 5. Add carry 1: $5 + 1 = 6$. Write 6.
    - Result: 62937

9. Multiplying by Numbers Ending in 1 (e.g., 23 x 31)

This is the reverse of the 'Vertical and Crosswise' method.

- **Trick**: For $AB \times CD$where $B = 1, D = 1$(e.g., $A1 \times C1$):
    1. Units digit: $1 \times 1 = 1$.
    2. Tens digit: $(A \times 1) + (C \times 1) = A + C$. (Carry over if sum is 2 digits).
    3. Hundreds digit: $A \times C$. (Add any carry-over).

- **Example**: 23×31
  - Units digit: $3 \times 1 = 3$
  - Tens digit: $(2 \times 1) + (3 \times 3) = 2 + 9 = 11$. Write 1, carry 1.
  - Hundreds digit: $2 \times 3 = 6$. Add carry 1: $6 + 1 = 7$.
  - Result: 713
  - So, $23 \times 31 = 713$

10. Vedic Multiplication for Any Two Numbers

This is the same as the 'Vertical and Crosswise' method but explained as a general principle:

- **Trick**: For any two numbers, say $XYZ$and ABC:
  1. Multiply units digits: $Z \times C$. Write the unit digit, carry over the tens digit.
  2. Multiply tens digits diagonally and add the units digits, plus carry-over: $(Y \times C) + (Z \times B) + (extcarry)$. Write the unit digit, carry over the tens digit.
  3. Multiply the outermost digits diagonally, sum the inner pair product, and add carry-over: $(X \times C) + (Z \times A) + (Y \times B) + (extcarry)$. Write the unit digit, carry over the tens digit.
  4. Continue this pattern, moving inwards for the left part of the multiplication.
- **Example**: 345×678
  - Units: $5 \times 8 = 40$. Write 0, carry 4.
  - Tens: $(4 \times 8) + (5 \times 7) + 4(extcarry) = 32 + 35 + 4 = 71$. Write 1, carry 7.
  - Hundreds: $(3 \times 8) + (5 \times 6) + (4 \times 7) + 7(extcarry) = 24 + 30 + 28 + 7 = 89$. Write 9, carry 8.
  - Thousands: $(3 \times 7) + (4 \times 6) + 8(extcarry) = 21 + 24 + 8 = 53$. Write 3, carry 5.
  - Ten Thousands: $(3 \times 6) + 5(extcarry) = 18 + 5 = 23$. Write 23.
  - Result: 233910
  - So, $345 \times 678 = 233910$

These 10 tricks cover a wide range of multiplication scenarios, enabling significant speed improvements.

Unit 2: Squaring and Cubing Tricks

11. Squaring Numbers Ending in 5

This is one of the most elegant and widely used Vedic Maths tricks.

- **Trick**: For a number $N5$(where $N$is the digit(s) before 5), the square is $(N \times (N+1))$followed by 25.
- **Example**: 35^2
  - The digit before 5 is 3. Calculate $3 \times (3+1) = 3 \times 4 = 12$.
  - Append 25 to 12. Result: 1225.
  - So, $35^2 = 1225$.
- **Example**: 125^2
  - The digits before 5 are 12. Calculate $12 \times (12+1) = 12 \times 13 = 156$.
  - Append 25 to 156. Result: 15625.
  - So, $125^2 = 15625$.

12. Squaring Numbers Near a Power of 10 (e.g., 98^2)

This method uses the Nikhilam Sutra concept.

- **Trick**: $(100-a)^2 = 10000 - 200a + a^2$
  The result is in two parts: the left part is $(100-2a)$and the right part is a^2. Ensure the right part has the correct number of digits (2 for base 100).
- **Example**: 98^2
  - $98$is $100-2$. So, $a = 2$.
  - Left part: $100 - 2a = 100 - 2(2) = 100 - 4 = 96$.
  - Right part: $a^2 = 2^2 = 4$. Pad with a zero to make it 2 digits: 04.
  - Combine: 9604.
  - So, $98^2 = 9604$.
- **Example**: 92^2
  - $92$is $100-8$. So, $a = 8$.
  - Left part: $100 - 2(8) = 100 - 16 = 84$.

    - Right part: $a^2 = 8^2 = 64$.
    - Combine: 8464.
    - So, $92^2 = 8464$.
- **Trick for numbers above 100**: $(100 + a)^2 = 10000 + 200a + a^2$
    - **Example**: 103^2
        - $103$is $100 + 3$. So, $a = 3$.
        - Left part: $100 + 2a = 100 + 2(3) = 100 + 6 = 106$.
        - Right part: $a^2 = 3^2 = 9$. Pad with a zero: 09.
        - Combine: 10609.
        - So, $103^2 = 10609$.

13. Squaring Numbers Using the $(A + B)^2$Identity

This is a general method that can be applied to any number by splitting it into two parts.

- **Trick**: $(A + B)^2 = A^2 + 2AB + B^2$. We can split a number, say XY, into $X0 + Y$or $X + Y$(if numbers are small).
- **Example**: 47^2
    - Split as $40 + 7$. Let $A = 40, B = 7$.
    - $A^2 = 40^2 = 1600$
    - $2AB = 2 \times 40 \times 7 = 80 \times 7 = 560$
    - $B^2 = 7^2 = 49$
    - Add them up: $1600 + 560 + 49 = 2160 + 49 = 2209$.
    - So, $47^2 = 2209$.
- **Example**: 23^2
    - Split as $20 + 3$. Let $A = 20, B = 3$.
    - $A^2 = 20^2 = 400$
    - $2AB = 2 \times 20 \times 3 = 40 \times 3 = 120$
    - $B^2 = 3^2 = 9$
    - Add them up: $400 + 120 + 9 = 529$.
    - So, $23^2 = 529$.

14. Squaring Numbers Using the $(A - B)^2$ Identity

Similar to the $(A + B)^2$ identity, useful when a number is close to a round number.

- **Trick**: $(A - B)^2 = A^2 - 2AB + B^2$.
- **Example**: 53^2
  - Split as $50 + 3$, use $(A + B)^2$: $50^2 + 2(50)(3) + 3^2 = 2500 + 300 + 9 = 2809$.
  - Alternatively, split as $60 - 7$. Let $A = 60, B = 7$.
  - $A^2 = 60^2 = 3600$
  - $2AB = 2 \times 60 \times 7 = 120 \times 7 = 840$
  - $B^2 = 7^2 = 49$
  - Result: $3600 - 840 + 49 = 2760 + 49 = 2809$.
  - So, $53^2 = 2809$.

15. Squaring Numbers Using $(X + Y)(X - Y) = X^2 - Y^2$

This trick helps in squaring numbers by relating them to known squares.

- **Trick**: $N^2 = (N + a)(N - a) + a^2$. Choose 'a' such that $(N+a)$ or $(N-a)$ is a round number (e.g., multiple of 10).
- **Example**: 46^2
  - Let $a = 4$. Then $N + a = 46 + 4 = 50$ and $N - a = 46 - 4 = 42$.
  - $46^2 = (50)(42) + 4^2$
  - $50 \times 42 = 5 \times 10 \times 42 = 5 \times 420 = 2100$.
  - $4^2 = 16$.
  - $46^2 = 2100 + 16 = 2116$.
  - So, $46^2 = 2116$.
- **Example**: 58^2
  - Let $a = 2$. Then $N + a = 58 + 2 = 60$ and $N - a = 58 - 2 = 56$.
  - $58^2 = (60)(56) + 2^2$
  - $60 \times 56 = 6 \times 560 = 3360$.
  - $2^2 = 4$.

- $58^2 = 3360 + 4 = 3364$.
- So, $58^2 = 3364$.

16. Squaring 3-Digit Numbers (General Method)

Can be done using the $(A+B)^2$or $(A-B)^2$method by splitting the number.

- **Example**: 123^2
  - Split as $100 + 23$. Let $A = 100, B = 23$.
  - $A^2 = 100^2 = 10000$
  - $2AB = 2 \times 100 \times 23 = 200 \times 23 = 4600$
  - $B^2 = 23^2$. We can use a previous trick: $23^2 = (20+3)^2 = 400 + 120 + 9 = 529$.
  - Add them up: $10000 + 4600 + 529 = 14600 + 529 = 15129$.
  - So, $123^2 = 15129$.
- **Example**: 256^2
  - Split as $250 + 6$. Let $A = 250, B = 6$.
  - $A^2 = 250^2 = (25imes10)^2 = 625imes100 = 62500$.
  - $2AB = 2 \times 250 \times 6 = 500 \times 6 = 3000$.
  - $B^2 = 6^2 = 36$.
  - Add them up: $62500 + 3000 + 36 = 65500 + 36 = 65536$.
  - So, $256^2 = 65536$.

17. Cubing Numbers Ending in 5

Similar to squaring numbers ending in 5, but with a slightly different rule for the preceding part.

- **Trick**: For a number $N5$, the cube is $(N \times (N+1))$followed by $N \times (N + 1) \times extfirstdigitofNimes3$? No, that's too complex. A simpler rule: For a number $N5$, the cube is calculated by $(N \times (N+1))$ and $(N \times (N + 1) \times extlastdigitofNimes3)$? No.

Let's use the structure $(Nimes(N+1))$and then the last three digits.

The formula is: $(N \times (N+1))$followed by $25 \times (N+1)$where the last digit of the first part is carried over.

A more structured approach:
For $N5^3$: Calculate $N(N+1)$and $N5$. The result is $(N(N+1))$and then $(N(N+1))$'s last digit $+25$. This is getting complicated. Let's try specific cases.

Correct Trick for $N5^3$: $(N \times (N+1))$followed by $25imes(N+1)$with carry-over.

Let's take an example:

- **Example**: 15^3
  - Here $N = 1. N + 1 = 2.$
  - First part: $N \times (N+1) = 1 \times 2 = 2.$
  - Second part: $25imes(N+1) = 25imes2 = 50.$ Since we need the last three digits for cube, 50will be 050. No, this is wrong.

Revisiting the $N5^3$**trick**: It's $(Nimes(N+1))$for the main part, and then the last digits are related to 25. This is often taught as: $N \times (N+1) \mid 25imes(N+1)$.
Example 15^3: $N = 1. N + 1 = 2. Nimes(N+1) = 2. 25imes(N+1) = 50$. Combine as $2 \mid 50$, giving 250. This is not 3375. This trick applies for $N25$, not $N5$. My apologies.

Correct approach for $N5^3$: Use the $(A+B)^3$expansion or a general multiplication method.

I8. Multiplication Using a Base (Nikhilam Sutra)

This method focuses on numbers close to powers of 10 (10, 100, 1000).

1. Base 10 (Both Below 10)

Method: Find differences from 10. Cross-subtract for the left part; multiply differences for the right part.

Example: 8 x 7 -> Differences: -2, -3. Left: 8 - 3 = 5. Right: (-2) x (-3) = 6. **Answer: 56**

2. Base 10 (Both Above 10)

Method: Same as above, but differences are positive. Cross-add.

Example: 12 x 14 -> Differences: +2, +4. Left: 12 + 4 = 16. Right: 2 x 4 = 8. **Answer: 168**

3. Base 100 (Both Below 100)

Method: Same logic. The right part must have two digits.

Example: 96 x 93 -> Differences: -4, -7. Left: 96 - 7 = 89. Right: (-4) x (-7) = 28. **Answer: 8928**

4. Base 100 (Both Above 100)

Method: Cross-add differences. Right part has two digits.

Example: 104 x 105 -> Differences: +4, +5. Left: 104 + 5 = 109. Right: 4 x 5 = 20. **Answer: 10920**

5. Base 100 (One Above, One Below)

Method: Cross-add/subtract. Add two zeros to the left part, then subtract the product of the differences.

Example: 104 x 97 -> Differences: +4, -3. Left: 104 - 3 = 101. Add zeros: 10100. Subtract (4 x 3 = 12). 10100 - 12 = **10088**

6. Base 1000 (Both Below)

Method: Right side must have three digits.

Example: 992 x 989 -> Differences: -8, -11. Left: 992 - 11 = 981. Right: 8 x 11 = 088. **Answer: 981088**

7. Base 1000 (Both Above)

Method: Cross-add. Right side gets three digits.

Example: 1006 x 1012 -> Left: 1006 + 12 = 1018. Right: 6 x 12 = 072. **Answer: 1018072**

8. Base 50 (Working Base Below)

Method: Base 50 is 100/2. Find differences from 50, cross-subtract, halve the left part, append the right part.

Example: 46 x 48 -> Differences: -4, -2. Left: (46 - 2)/2 = 22. Right: 4 x 2 = 08. **Answer: 2208**

9. Base 50 (Working Base Above)

Method: Same as above, but cross-add before halving.

Example: 52 x 54 -> Differences: +2, +4. Left: (52 + 4)/2 = 28. Right: 2 x 4 = 08. **Answer: 2808**

10. Base 200 (Working Base)

Method: Base 200 is 100 x 2. Cross-add/subtract, multiply the left part by 2.

Example: 204 x 206 -> Left: (204 + 6) x 2 = 420. Right: 4 x 6 = 24. **Answer: 42024**

I9. Special Multiplication Cases

1. Multiplying by 11 (2-Digit)

Method: Split the digits and insert their sum in the middle. (Carry over if sum > 9).

Example: 45 x 11 -> 4 _ 5 -> 4 + 5 = 9. **Answer: 495**

2. Multiplying by 11 (Carry Over)

Method: When the sum is 10 or more, carry the 1 to the left digit.

Example: 78 x 11 -> 7 _ 8 -> 7 + 8 = 15. Carry 1 to 7 -> 858. **Answer: 858**

3. Multiplying by 11 (Large Numbers)

Method: Write the rightmost digit. Add adjacent pairs moving left. Write the leftmost digit.

Example: 234 x 11 -> 4, (4+3)=7, (3+2)=5, 2. **Answer: 2574**

4. Multiplying by 111

Method: Similar to 11, but add up to three adjacent digits at a time.

Example: 234 x 111 -> 4, (4+3)=7, (4+3+2)=9, (3+2)=5, 2. **Answer: 25974**

20. Multiplying by 99 (Equal Digits)

Method: Subtract 1 from the multiplicand (Left part). Subtract that result from 99 (Right part).

Example: 46 x 99 -> Left: 46 - 1 = 45. Right: 99 - 45 = 54. **Answer: 4554**

1. Multiplying by 999 (Equal Digits)

Method: Same as above, scaled up.

Example: 456 x 999 -> Left: 456 - 1 = 455. Right: 999 - 455 = 544. **Answer: 455544**

2. Multiplying by 9s (More 9s than Digits)

Method: Add leading zeros to the multiplicand, then apply the same rule.

Example: 45 x 999 -> Treat as 045 x 999. Left: 045 - 1 = 044. Right: 999 - 044 = 955. **Answer: 44955**

3. Multiplying by 9s (Fewer 9s than Digits)

Method: Add zeros equal to the number of 9s, then subtract the original number.

Example: 243 x 99 -> 24300 - 243 = **24057**

4. Tens Digit Same, Units Add to 10

Method: Multiply the tens digit by (tens digit + 1). Multiply the units digits.

Example: 63 x 67 -> Left: 6 x (6+1) = 42. Right: 3 x 7 = 21. **Answer: 4221**

5. Units Digit Same, Tens Add to 10

Method: Multiply the tens digits and add the units digit. Square the units digit (two digits).

Example: 43 x 63 -> Left: (4 x 6) + 3 = 27. Right: 3 x 3 = 09. **Answer: 2709**

21. Multiplying by 5

Method: Divide the number by 2 and multiply by 10 (or append 0).

Example: 428 x 5 -> 428 / 2 = 214. **Answer: 2140**

1. Multiplying by 25

Method: Divide the number by 4 and multiply by 100 (append 00).

Example: 124 x 25 -> 124 / 4 = 31. **Answer: 3100**

2. Multiplying by 125

Method: Divide by 8 and multiply by 1000.

Example: 64 x 125 -> 64 / 8 = 8. **Answer: 8000**

22. 2-Digit Cross Multiplication (Urdhva Tiryagbhyam)

Method: Vertical right, cross-add middle, vertical left. (Carry over at each step).

Example: 21 x 32 -> Right: 1x2 = 2. Middle: (2x2) + (1x3) = 7. Left: 2x3 = 6. **Answer: 672**

23. 3-Digit Cross Multiplication

Method: Applies the vertical/crosswise pattern across 5 steps (Right vertical, right cross, full cross, left cross, left vertical).

Example: 121 x 112 -> Step 1: 1x2 = 2. Step 2: (2x2)+(1x1) = 5. Step 3: (1x2)+(1x1)+(2x1) = 5. Step 4: (1x1)+(2x1) = 3. Step 5: 1x1 = 1. **Answer: 13552**

24. Squaring Techniques

1. Squaring Numbers Ending in 5

Method: Multiply the first digit by itself plus one. Append 25.

Example: 65 x 65 -> Left: 6 x (6+1) = 42. Right: 25. **Answer: 4225**

2. Squaring Near 50 (Above)

Method: Base is 25. Add the difference from 50 to 25. Square the difference for the right side (two digits).

Example: 53 x 53 -> Difference is 3. Left: 25 + 3 = 28. Right: 3^2 = 09. **Answer: 2809**

3. Squaring Near 50 (Below)

Method: Subtract the difference from 25. Square the difference.

Example: 48 x 48 -> Difference is 2. Left: 25 - 2 = 23. Right: 2^2 = 04. **Answer: 2304**

4. Squaring Near 100 (Below)

Method: Subtract the difference from the number itself. Square the difference.

Example: 96 x 96 -> Difference is 4. Left: 96 - 4 = 92. Right: 4^2 = 16. **Answer: 9216**

5. Squaring Near 100 (Above)

Method: Add the difference to the number itself. Square the difference.

Example: 107 x 107 -> Difference is 7. Left: 107 + 7 = 114. Right: 7^2 = 49. **Answer: 11449**

6. The Duplex Method (2-Digit Squares)

Method: Uses formula a^2 | 2ab | b^2, calculating from right to left with carryovers.

Example: 34^2 -> a=3, b=4. Right (b^2): 16 (write 6, carry 1). Middle (2ab): 2x3x4 = 24 + 1(carry) = 25 (write 5, carry 2). Left (a^2): 9 + 2(carry) = 11. **Answer: 1156**

7. Squaring Series of 1s (Magic Palindromes)

Method: Count the number of 1s. Count up to that number and back down to 1.

Example: 1111 x 1111 (Four 1s). **Answer: 1234321**

8. Squaring Series of 3s

Method: Uses the pattern 1089. For each additional 3, add a 1 and an 8.

Example: 333^2 -> Three digits, so write two 1s, one 0, two 8s, one 9. **Answer: 110889**

9. Squaring Series of 6s

Method: Uses the pattern 4356. For each additional 6, add a 4 and a 5.

Example: 666^2 -> Two 4s, one 3, two 5s, one 6. **Answer: 443556**

10. Squaring Series of 9s

Method: Uses the pattern 9801. For each additional 9, add a 9 and a 0.

Example: 999^2 -> Two 9s, one 8, two 0s, one 1. **Answer: 998001**

25. Cubes and Roots

1 Cubing Numbers Near 10 (Above)

Method: Add 2x difference to the number. (Difference^2 x 3) for middle. (Difference^3) for right.

Example: 12^3 -> Diff = 2. Left: 12 + 4 = 16. Mid: 2^2 x 3 = 12 (write 2, carry 1). Right: 2^3 = 8. Left becomes 17. **Answer: 1728**

2. Cubing Numbers Ending in 1

Method: Ratio method. Write 1, then the tens digit (x), then x^2, then x^3. Double the middle two and add.

Example: 21^3 -> Write: 8, 4, 2, 1. Double middle: 8, 4. Add vertically: (8), (4+8=12, carry 1), (2+4=6), (1). Left becomes 9. **Answer: 9261**

3. Square Roots of Perfect Squares

Method: Look at the last digit to find the two possible ending digits (e.g., ends in 4 -> root ends in 2 or 8). Group the last two digits, find the nearest square for the rest. Check the middle number ending in 5 to verify.

Example: √3136 -> Ends in 6, so root ends in 4 or 6. First part is near 25 (5^2). Root is 54 or 56. 55^2 is 3025. 3136 > 3025, so **Answer: 56**

4. Cube Roots of Perfect Cubes (Up to 6 Digits)

Method: The last digit of the cube uniquely maps to the last digit of the root (e.g., 2<->8, 3<->7, others stay the same). Ignore the last 3 digits. Find the closest smaller cube for the remaining numbers.

Example: $\sqrt[3]{103823}$ -> Ends in 3, so root ends in 7. Strike last three (823). Remaining is 103. Closest smaller cube is 64 (4^3). **Answer: 47**

5. Estimating Non-Perfect Square Roots

Method: Use formula: √S ≈ √N + (S - N) / (2 * √N), where N is the nearest perfect square.

Example: √28 -> Nearest square N=25 (root is 5). 5 + (28 - 25) / (2 * 5) = 5 + 3/10 = 5.3. **Answer: ~5.29**

26. Division Strategies

1. Dividing by 5

Method: Multiply the number by 2 and divide by 10 (move decimal one place left).

Example: 134 / 5 -> 134 x 2 = 268. Move decimal. **Answer: 26.8**

2. Dividing by 25

Method: Multiply by 4 and divide by 100.

Example: 134 / 25 -> 134 x 4 = 536. Move decimal two places. **Answer: 5.36**

3. Dividing by 125

Method: Multiply by 8 and divide by 1000.

Example: 210 / 125 -> 210 x 8 = 1680. Move decimal three places. **Answer: 1.68**

4. Dividing by 9 (Single Digit Quotient)

Method: The first digit of the dividend is the quotient. The sum of the digits is the remainder.

Example: 71 / 9 -> Quotient = 7. Remainder = 7 + 1 = 8. **Answer: 7 R 8**

5. Dividing by 9 (Multi-Digit)

Method: Bring down the first digit. Add it to the next digit to get the next part of the quotient, and so on. The final sum is the remainder.

Example: 214 / 9 -> Q1 = 2. Q2 = 2+1 = 3. Remainder = 3+4 = 7. **Answer: 23 R 7**

6. Dividing by 99

Method: Split dividend into pairs from the right. Add the pairs together.

Example: 1425 / 99 -> Split as 14 | 25. Quotient = 14. Remainder = 14 + 25 = 39. **Answer: 14 R 39**

VI. Addition, Subtraction & Verification

7. Subtraction from a Base (All from 9, Last from 10)

Method: To subtract a number from 100, 1000, 10000, etc., subtract every digit from 9, except the last non-zero digit, which you subtract from 10.

Example: 1000 - 456 -> (9-4)=5, (9-5)=4, (10-6)=4. **Answer: 544**

8. Fraction Addition (Cross Multiplication)

Method: (a/b) + (c/d) = (ad + bc) / bd. Do this mentally without finding the LCM for fast estimates.

Example: 2/3 + 4/5 -> Top: (2x5) + (3x4) = 10 + 12 = 22. Bottom: 3x5 = 15. **Answer: 22/15**

9. Divisibility by 7 (Osculation)

Method: Double the last digit and subtract it from the remaining truncated number. If the result is divisible by 7, the whole number is.

Example: 343 -> 3 x 2 = 6. 34 - 6 = 28. (28 is divisible by 7). **Answer: Yes**

10. The Digit Sum Check (Navasesh / Casting out 9s)

Method: To verify any calculation, sum the digits of the operands to a single digit (ignoring 9s). The mathematical operation performed on these single digits must equal the digit sum of the answer.

Example: Verify 43 x 47 = 2021.

Digit sum of 43 -> 4+3 = 7.

Digit sum of 47 -> 4+7 = 11 -> 1+1 = 2.

7 x 2 = 14 -> 1+4 = 5.

Digit sum of 2021 -> 2+0+2+1 = 5. **Answer: The calculation is verified (5 = 5).**

Will continue, follow us....